HYBRID QUANTUM-CLASSICAL VERSUS CLASSICAL DEEP LEARNING FOR FOUR-CLASS ALZHEIMER'S MRI CLASSIFICATION

A PROJECT REPORT

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BONAFIDE CERTIFICATE

This is to certify that this project report entitled **"Hybrid Quantum-Classical versus Classical Deep Learning for Four-Class Alzheimer's MRI Classification"** is the bonafide work of **"Reddi Abhinaya Sai (CH.SC.U4AIE24061) and Rayankula Nidhi (CH.SC.U4AIE24085)"**, who carried out the project work under my supervision.

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ABSTRACT

This project compares classical and hybrid quantum-classical models for four-class Alzheimer MRI classification. The dataset contains NonDemented, VeryMildDemented, MildDemented and ModerateDemented images. Both model families use an ImageNet-pretrained ResNet18 encoder. The classical models adapt the encoder and classification head to MRI data, while the hybrid models use its frozen 512-dimensional features. An eight-dimensional projection supplies rotation angles to an 8-qubit, 2-layer variational circuit simulated with PennyLane. Hybrid Quantum Model 1 uses the circuit pathway alone; Hybrid C adds a parallel classical branch. StratifiedGroupKFold is used in the updated classical evaluation to reduce leakage between related samples while maintaining approximate class balance across folds.

The classical head achieves 94.69% test accuracy on the augmented dataset and 96.78% on the original dataset. The hybrid results are 85.83% for the circuit-only head and 88.33% for Hybrid C, with macro-F1 of 0.8821 for the latter. The augmented classical result is 6.36 percentage points above Hybrid C and 8.86 points above the circuit-only model. The original-data classical head gives the strongest result. The experiments show that the variational circuit can be trained as part of an MRI classifier, while the classical heads provide higher accuracy in this study.

Keywords: Alzheimer's disease, MRI classification, ResNet18, hybrid quantum-classical neural network, variational quantum circuit, transfer learning

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LIST OF SYMBOLS AND ABBREVIATIONS

AD	- Alzheimer's Disease
MRI	- Magnetic Resonance Imaging
CNN	- Convolutional Neural Network
ResNet	- Residual Network
VQC	- Variational Quantum Circuit
NISQ	- Noisy Intermediate-Scale Quantum
CNOT	- Controlled-NOT gate
ReLU	- Rectified Linear Unit
SVM	- Support Vector Machine
RBF	- Radial Basis Function (SVM kernel)
AdamW	- Adam optimiser with decoupled weight decay
AUC	- Area Under the ROC Curve (one-vs-rest, weighted)
CI	- Confidence Interval (95% Wilson)
GPU / CPU	- Graphics Processing Unit / Central Processing Unit
Cfg	- Data configuration (A, B or C)
Hybrid C	- Best hybrid model: circuit plus classical branch
f	- 512-dimensional feature vector from ResNet18
z	- Vector of eight rotation angles, one per qubit
θ	- Trainable weights of the quantum circuit
o_k	- Pauli-Z expectation value of qubit k
$\hat{y}$	- Output class scores of the classifier
λ	- Learned mixing weight between circuit and classical branch
n, L	- Number of qubits, number of circuit layers
k, n	- Correct predictions, test-set size (accuracy k/n)

CHAPTER 1

INTRODUCTION

1.1. BACKGROUND

Alzheimer's disease is a progressive neurodegenerative disorder that affects memory, reasoning and everyday functioning. Identifying the stage of impairment helps clinicians plan follow-up and helps families prepare for changing care needs. A four-class task is useful for studying this progression because it distinguishes non-demented, very mild, mild and moderate categories rather than reducing the problem to a binary label.

Structural MRI provides information about tissue loss and anatomical changes associated with dementia. Medial temporal atrophy, cortical thinning and ventricular enlargement are relevant patterns, although early changes can overlap with normal ageing. Reviewing these patterns consistently is difficult, especially in very mild cases. Image classification models offer a way to study whether learned features can support a consistent distinction between the supplied severity categories.

1.2. MOTIVATION

Convolutional neural networks (CNNs) trained by transfer learning are the standard approach to medical image classification. They work well, but they carry millions of trainable parameters. A different idea has gained attention in recent years: hybrid quantum-classical networks, in which a small parameterised quantum circuit acts as a trainable non-linear layer between classical layers. The idea was set out for transfer learning by Mari et al. [10] and has since been tried on MRI-based dementia detection [4, 7]. The appeal is compactness, since a circuit with a few dozen angles can sit inside a classifier whose trainable head is only a few thousand parameters.

We use classical simulation to study a variational circuit as a trainable classification layer. The main question is how well that layer performs when attached to a strong MRI feature extractor. Mari et al. [10] provide the transfer-learning foundation, while Bowles et al. [5] motivate careful selection of classical baselines. We therefore consider accuracy, class-wise performance and computational cost together.

1.3. RESEARCH QUESTIONS AND OBJECTIVES

The project addresses two research questions. Q1: how does an 8-qubit, 2-layer variational quantum circuit perform relative to a fine-tuned ResNet18 on four-class

Alzheimer MRI classification, and what are its parameter and inference costs? Q2: what additional evidence is needed to distinguish the contribution of the circuit from that of the classical components? The reported results address Q1 descriptively, while the evaluation protocol in Section 4.6 defines the controls required for Q2.

To answer them, the project set out the following objectives:

- Build two fine-tuned ResNet18 baselines: Classical Model 1 on a balanced 6,000-image augmented subset and Classical Model 2 on the 6,400-image original folder.
- Freeze the fine-tuned network as a feature extractor and attach 8-qubit, 2-layer VQC heads (Hybrid Quantum Model 1 and Hybrid C), compared with the classical model on the same test images.
- Compare accuracy, macro-F1, per-class F1, uncertainty intervals, trainable parameters and recorded inference time.
- Audit the results with integrity checks for duplicates and for the independence of the validation set.
- Identify the controls needed to isolate the circuit contribution and strengthen the evaluation.

1.4. CONTRIBUTIONS

The project applies quantum transfer learning to four-class MRI classification. Its main contributions are:

- A reproducible description of two classical classifiers and two principal hybrid classifiers, including their data configurations, architecture and recorded training settings.
- A comparison of classical-head accuracies of 94.69% on augmented data and 96.78% on original data with hybrid accuracies of 85.83% and 88.33%.
- An experimental account of how circuit width, projection, training settings and a parallel classical branch affect four-class performance.
- Use of StratifiedGroupKFold in the updated classical evaluation to reduce leakage between grouped samples, supported by a separate audit of the earlier experiments.

1.5. SCOPE AND DELIMITATIONS

The study uses public 2-D MRI images and compares ResNet18 classifiers with small simulated variational circuits. The earlier experiments use balanced augmented subsets and the complete original folder. The updated classical evaluation uses StratifiedGroupKFold. Metrics describe image classification on the supplied dataset, and the quantum component runs on a noiseless state-vector simulator. Section 5.7 brings together the conditions relevant to interpreting these results.

1.6. ORGANISATION OF THE REPORT

Chapter 2 reviews structural MRI, transfer learning and variational quantum circuits. Chapter 3 defines the classification task, dataset and evaluation measures. Chapter 4 presents the model architectures, implementation and experimental protocol. Chapter 5 compares the results and analyses their interpretation. Chapter 6 summarises the findings and identifies the next research steps.

CHAPTER 2

LITERATURE REVIEW AND PROBLEM BACKGROUND

2.1. ALZHEIMER'S DISEASE AND STRUCTURAL MRI

2.1.1. Clinical Stages

The dataset assigns each slice to NonDemented, VeryMildDemented, MildDemented or ModerateDemented. These categories provide a four-class learning target spanning normal status and increasing dementia severity. In this report they are treated as the supplied dataset labels. The distinction between NonDemented and VeryMildDemented is particularly relevant to the class-wise evaluation because subtle structural changes can overlap across these categories.

2.1.2. What the Scans Show

Structural MRI reveals disease-associated changes through tissue volume and anatomical shape. Relevant changes include medial temporal atrophy, cortical thinning and ventricular enlargement. Early changes can be subtle and may overlap with normal ageing, which makes learned image features useful for a classification study. The benchmark used here represents scans as individual 2-D JPEG slices.

2.2. DEEP LEARNING FOR ALZHEIMER'S CLASSIFICATION

Convolutional networks learn their own features. A convolution layer slides small filters across the image, early layers respond to edges and textures, and deeper layers combine these into task-specific patterns. Training one from scratch needs a lot of labelled data, which medical imaging rarely offers, so the usual route is to start from a network pretrained on a large natural-image collection and adapt it.

Wen et al. [18] discuss reproducible evaluation in MRI classification and the importance of separating related samples across partitions. We address this issue with group-aware folding in the updated classical evaluation. Stratification aims to preserve the class distribution, while the grouping constraint keeps samples with the same group identifier in one fold. This is particularly relevant when several images are derived from the same source.

2.3. TRANSFER LEARNING WITH RESNET18

2.3.1. Residual Learning

ResNet18 is an 18-layer residual network introduced by He et al. [6]. Very deep plain networks are hard to optimise, and residual networks address this by letting each block learn a correction to its input instead of a whole new mapping. If x is the input to a block and F is the mapping learned by its convolution layers, the block outputs

$$y = F(x, \{Wi\}) + x \tag{2.1}$$

The addition is carried by an identity shortcut (Figure 2.1), which gives gradients a direct path back through the network. ResNet18 stacks such blocks in four stages, summarised in Table 2.1, and ends with global average pooling that turns the last feature map into a 512-dimensional vector. That vector is the interface between the classical encoder and the quantum head in our hybrid models.

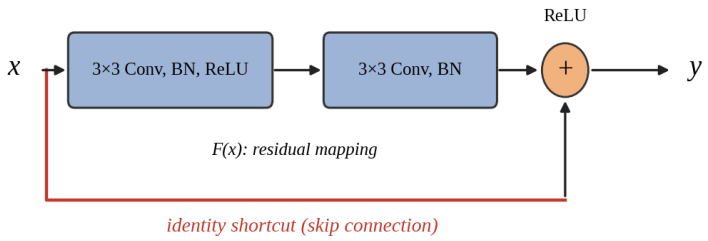


Figure 2.1: A basic residual block. The identity shortcut adds the block input to the output of two convolution layers.

Table 2.1: Stages of ResNet18 and what each contributes to this project

Stage	Composition	Role in this project
conv1 + pooling	7×7 convolution, batch norm, ReLU, max pooling	Generic edge and texture filters; kept frozen
layer1, layer2	Two basic blocks each, 64 and 128 channels	Early features; kept frozen
layer3	Two basic blocks, 256 channels	Fine-tuned with a small learning rate
layer4	Two basic blocks, 512 channels	Fine-tuned with a small learning rate
Global average pooling	Averages each of the 512 channels	Produces the 512-dimensional feature vector
Classifier	Dropout and a 512→4 linear layer	Replaced and trained with a larger learning rate; removed for the hybrid heads

2.3.2. Fine-Tuning Strategy

Pretrained on ImageNet, the early layers of ResNet18 capture generic edges and textures, while the later layers (layer3 and layer4) capture task-specific structure. Freezing the early layers and fine-tuning the later ones is a common and data-efficient way to adapt the network. Using smaller learning rates for the pretrained layers than for the new classifier keeps the useful pretrained filters from being overwritten. We follow this recipe in Chapter 4.

2.4. QUANTUM COMPUTING FUNDAMENTALS

2.4.1. Qubits, Gates and Measurement

A classical bit is either 0 or 1. A qubit is described by a state that can be a superposition of the two basis states,

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle , \ |\alpha|^2 + |\beta|^2 = 1 \tag{2.2}$$

where α and β are complex amplitudes. A register of n qubits is described by 2^n amplitudes, so the 8-qubit register used here has 256. Operations on qubits are unitary gates. Single-qubit rotation gates turn the state about an axis by a chosen angle, for example

$$RX(\varphi) = \exp(-i\,\varphi\, X / 2) \tag{2.3}$$

where X is the Pauli-X matrix, and entangling gates such as the controlled-NOT (CNOT) link the states of two qubits so that they can no longer be described separately. Reading out the circuit means measuring an observable. For the Pauli-Z observable on one qubit the expectation value is

$$\langle Z\rangle = |\alpha|^2 - |\beta|^2 \in [-1, 1] \tag{2.4}$$

which is a smooth function of the rotation angles that were applied. This smoothness is what allows a circuit to be trained by gradient methods.

2.4.2. The NISQ Era and Classical Simulation

The NISQ framework described by Preskill [13] motivates hybrid algorithms that combine classical processing with small parameterised circuits. In this project, an 8-qubit state is represented by 256 complex amplitudes and evolved on a classical state-vector simulator. Exact expectation values provide a differentiable output for training. This implementation evaluates circuit-based learning without introducing finite-shot measurement variability.

2.5. VARIATIONAL QUANTUM CIRCUITS AS NEURAL LAYERS

2.5.1. Structure of the Circuit

A variational quantum circuit encodes a classical vector into qubit rotation angles, applies trainable entangling gates and returns measured expectation values [2, 14]. The circuit used here has three parts, detailed in Section 4.4. First, angle embedding sets one single-qubit rotation per input feature. Second, StronglyEntanglingLayers applies, in each of L layers, three trainable rotations to every qubit followed by a ring of entangling gates, which gives $L \times n \times 3$ weights, that is 48 for $n = 8$ qubits and $L = 2$. Third, the Pauli-Z expectation value of every qubit is measured, giving an n-dimensional vector in $[-1, 1]$ that is passed to a classical head.

2.5.2. Training Inside a Classical Network

Because the measured values are differentiable with respect to both the inputs and the circuit weights, the circuit can sit inside a larger model and be trained end to end. PennyLane [3] provides this differentiable circuit interface and connects to PyTorch [12], so the loss gradient flows through the circuit into the classical layers before it, by backpropagation or by the adjoint method. This is what makes the hybrid model easy to build and also what makes its comparison with a classical layer natural, since the two occupy the same place in the network.

2.5.3. Trainability

Circuit depth, width and initialisation influence optimisation. Barren plateaus describe regimes in which gradients become very small as a circuit grows [11]. The chosen 8-qubit, 2-layer design keeps the state representation and parameter count manageable. Gradient-based training is feasible for this design; the experiments do not separately diagnose barren plateaus or attribute unsuccessful runs to them.

2.6. RELATED WORK

Quantum transfer learning attaches a small VQC to a pretrained classical network that is used as a fixed feature extractor [10]. For dementia detection from MRI, Kim reports a hybrid network with a VQC that outperformed its classical counterpart on a two-class demented versus non-demented task [7]. Bhowmik et al. apply quantum transfer learning to binary classification on OASIS-2 and report gains over a deliberately weak classical base model with far fewer parameters [4]. Table 2.2 sets these works beside the two methodological studies that most influenced our design.

Table 2.2: Summary of related work and the position of this project

Work	Task or topic	Approach	Relevance here
Mari et al. [10]	Transfer learning	Pretrained network as fixed feature extractor with a VQC on top	Origin of the design we use
Kim [7]	Two-class dementia detection from MRI	Hybrid network with a VQC	Reports a binary hybrid comparison
Bhowmik et al. [4]	Binary classification, OASIS-2	Quantum transfer learning	Examines binary quantum transfer learning
Wen et al. [18]	CNNs for AD classification	Overview and reproducible evaluation	Warns against image-level splits
Bowles et al. [5]	Benchmarking quantum ML	Quantum models against classical baselines	Careful baselines remove many apparent advantages
This project	Four-class AD grading	Two classical and two hybrid models, audit and controlled protocol	Extends binary studies to graded stages; evaluative, not a new architecture

2.7. RESEARCH GAP AND POSITIONING

The reviewed work motivates two directions: extending binary classification to four severity categories and testing hybrid heads against a strong classical encoder. We combine these directions through a ResNet18 feature pipeline and an 8-qubit classifier. Group-aware evaluation is also part of the design, since repeated or related images can make a conventional image-level split overly optimistic. The contribution is an empirical comparison of these components rather than a new circuit architecture.

CHAPTER 3

PROBLEM STATEMENT AND METHODOLOGY

3.1. PROBLEM STATEMENT

Given a single 2-D structural MRI slice, the task is to assign it to one of four classes: NonDemented, VeryMildDemented, MildDemented or ModerateDemented. Each input is an image resized to 224 × 224 pixels with three channels and the output is one of four labels. We learn the mapping with two kinds of classifier built on the same ResNet18: a classical classifier, and a hybrid classifier whose circuit-based head is trained on the frozen ResNet18 features.

The analysis considers predictive performance and computational cost. The shared Configuration B test set permits a direct descriptive comparison of the principal classical and hybrid models. Configuration C provides an additional classical result under the original class distribution. The contribution of individual components is addressed through the proposed ablation protocol in Chapter 4.

3.2. DATASET

We use the Augmented Alzheimer MRI Dataset from Kaggle (uraninjo) [1]. It ships in two folders with the same four classes of 2-D MRI slices stored as JPEG files. The Original folder holds 6,400 images and is heavily imbalanced: it has only 64 ModerateDemented images against 3,200 NonDemented ones. The Augmented folder, derived from the original set by image transformations, holds 33,984 images and is far more balanced. Table 3.1 gives the counts, and Figure 3.1 shows them on a logarithmic axis so that the small classes remain visible.

Table 3.1: Images per class in the Original and Augmented folders

Class	Original folder	Augmented folder
NonDemented	3,200	9,600
VeryMildDemented	2,240	8,960
MildDemented	896	8,960
ModerateDemented	64	6,464
Total	6,400	33,984

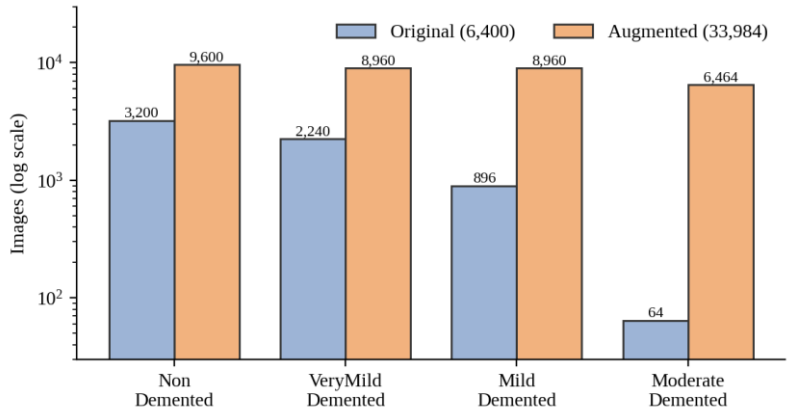


Figure 3.1: Class distribution of the Original and Augmented folders (logarithmic axis).

Augmentation changes the number of available training examples and the apparent class balance, but does not add independent source scans. For example, the ModerateDemented category increases from 64 original images to 6,464 augmented images. This distinction motivates reporting both folder counts and the configuration used for each result.

3.3. DATA CONFIGURATIONS

The earlier experiments use three configurations, summarised in Table 3.2. Configurations A and B contain equal class counts and use 80/10/10 train, validation and test partitions. Configuration C retains the original class proportions and uses a 70/15/15 partition. These development splits use seed 42. The updated classical runs instead use StratifiedGroupKFold to reduce leakage. Samples carrying the same group identifier remain in the same fold, while the algorithm seeks to preserve the class proportions. The split counts in Table 3.2 describe the earlier experiments, not a reconstruction of the updated folds.

Table 3.2: Data configurations used in the experiments

Cfg	Source and sampling	Images	Earlier train / val / test	Used for
A	Augmented folder, 1,000 per class	4,000	3,200 / 400 / 400	Notebook 1: classical baselines, first quantum pilot
B	Augmented folder, 1,500 per class	6,000	4,800 / 600 / 600	Notebook 2: main classical vs hybrid comparison
C	Original folder, all images	6,400	4,480 / 960 / 960	Notebook 1: best classical model (Classical Model 2)

Grouping and stratification serve different purposes. Stratification helps keep the class distribution similar across folds, while grouping keeps samples with the same assigned

identifier together. This is useful for evaluating an augmented dataset, where related images should be treated as a group rather than as independent examples.

3.4. METHODOLOGICAL FLOW

Figure 3.2 summarises the workflow. The encoder is trained first, after which its frozen features are used to fit the hybrid heads. The updated classical evaluation uses StratifiedGroupKFold; the earlier hybrid development runs and their audit are reported separately.

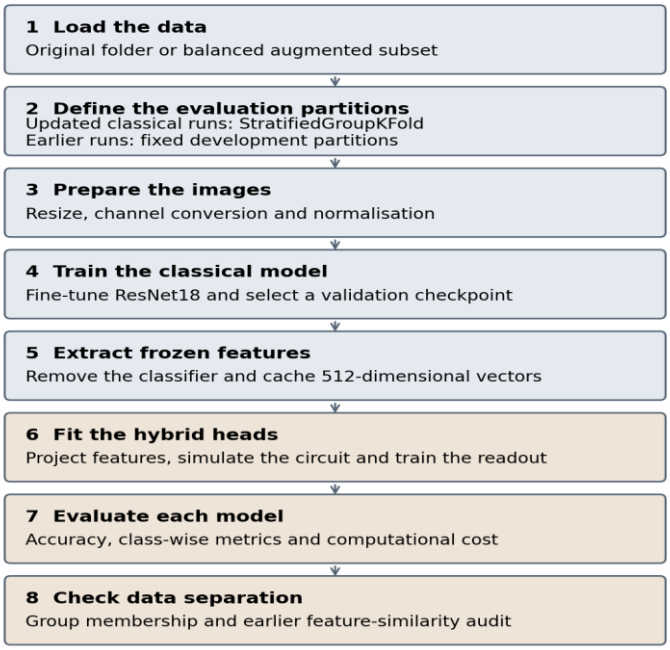


Figure 3.2: Methodological flow of the study, from data loading to integrity checks.

The eight stages in order:

1. Load and sample. Read the Kaggle folders. For Configurations A and B, draw an equal number of images per class from the Augmented folder; for Configuration C use every original image.
2. Partition the data. Use StratifiedGroupKFold for the updated classical evaluation, keeping each assigned group within one fold and aiming for similar class proportions. The earlier development experiments used the fixed partitions in Table 3.2.
3. Pre-process. Resize to 224×224, give each slice three channels and normalise with ImageNet statistics. Training images receive random flips; Configuration C adds small rotations and brightness and contrast jitter.

4. Train the CNN. Fine-tune ImageNet-pretrained ResNet18 (and EfficientNet-B0 in Notebook 1) with AdamW, separate learning rates for pretrained blocks and the new head, and early stopping on validation accuracy. This gives the classical results.
5. Extract features. Freeze the best ResNet18, drop its classifier and store the 512-number vector of every image.
6. Train the shallow and quantum heads. An SVM and the hybrid circuit models each learn from the stored vectors. The hybrids standardise them with training statistics, compress them to eight angles, run the 8-qubit circuit and read out four class scores.
7. Evaluate on the test images. Accuracy, macro and weighted F1, per-class precision and recall, confusion matrices, one-vs-rest AUC, parameter counts and inference time.
8. Check the partitions. Examine group separation and search for repeated or highly similar samples. Retain the earlier feature-similarity audit as part of the experimental record.

Implementation uses PyTorch 2.11 and PennyLane 0.45.1. CNN training runs on a Tesla T4 GPU, and the quantum layer runs on a state-vector simulator. Earlier development runs used time-limited training passes. Caching the encoder features makes it practical to train several small heads without repeatedly applying the convolutional network.

3.5. PRE-PROCESSING

Images are resized to 224 × 224 pixels, represented with three channels and normalised using ImageNet statistics. Classical Model 1 explicitly converts each image to grayscale before producing the three-channel input. Training applies horizontal flips; Configuration C also applies small rotations and brightness/contrast jitter. Validation and test transformations are deterministic. Figure 3.3 distinguishes the image preparation and feature preparation stages, including the use of training statistics for feature standardisation.

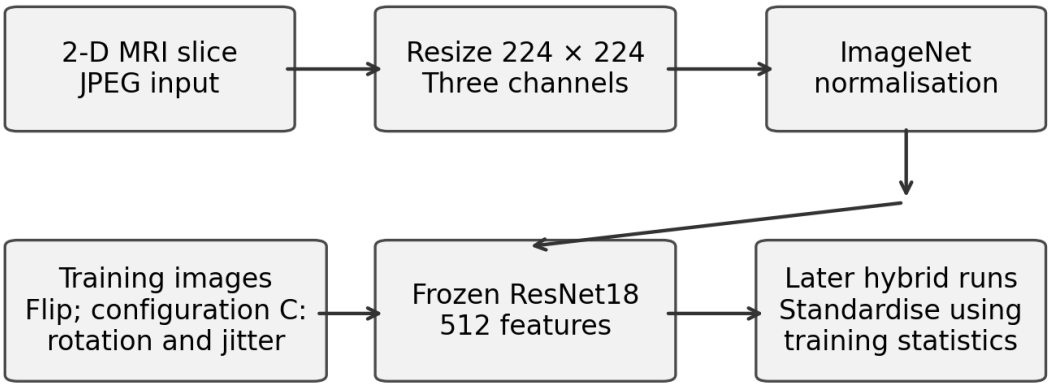


Figure 3.3: Image and feature preparation for the classical and hybrid classifiers.

3.6. EVALUATION METRICS

We report accuracy, macro and weighted F1, per-class precision, recall and F1, confusion matrices, one-vs-rest (OvR) weighted ROC-AUC, trainable parameter counts and inference time. Accuracy is the fraction of test images classified correctly, k out of n:

$$\text{Accuracy} = k / n \tag{3.1}$$

For each class, precision is the fraction of images predicted as that class that really belong to it, and recall is the fraction of images of that class that were found. F1 is their harmonic mean:

$$\text{Precision} = TP / (TP + FP), \quad \text{Recall} = TP / (TP + FN) \tag{3.2}$$

$$F1 = 2 \cdot \text{Precision} \cdot \text{Recall} / (\text{Precision} + \text{Recall}) \tag{3.3}$$

Macro-F1 averages the four class scores with equal weight; weighted F1 weights them by class support. A nominal 95% Wilson interval can summarise finite-test-set uncertainty when the correct-prediction count and test-set size are available. For observed accuracy p, test-set size n and u = 1.96, the interval is:

$$[\, p + u^2/2n \pm u \sqrt{(\, p(1-p)/n + u^2/4n^2 \,)} \,] / (\, 1 + u^2/n \,) \tag{3.4}$$

For Hybrid C, 530 correct classifications out of 600 give a nominal Wilson interval of approximately 85.5–90.7%. This is an interval for one accuracy, not a test of the difference between two models. A paired comparison requires the individual predictions. The updated classical percentages are reported without reconstructed counts or intervals.

CHAPTER 4

SYSTEM DESIGN AND EXPERIMENTAL WORK

4.1. OVERVIEW OF THE SYSTEM AND THE MODELS COMPARED

The system has two parts that are trained separately. The first is a classical encoder: an ImageNet-pretrained ResNet18 that is fine-tuned on the MRI data. After fine-tuning, its classification layer is removed and the network becomes a frozen feature extractor. The second part is the head, which maps the 512-dimensional feature vector to four class scores. A classical model is the fine-tuned ResNet18 with its own linear classifier, trained end to end on images. A hybrid quantum model keeps the fine-tuned ResNet18 frozen and trains a small head, built around an 8-qubit variational circuit, on the cached features. The expensive convolutional computation is therefore done once by the classical network, and the circuit is asked to do only the final, compact part of the classification. Table 4.1 lists the four models that are compared throughout the report.

Table 4.1: The four models compared

Model	What it contains	Data and test images	Test accuracy
Classical Model 1	ImageNet-pretrained ResNet18; layer3, layer4 and a new 512→4 head fine-tuned end to end; grayscale 3-channel input, dropout 0.3, label smoothing 0.1; early stopping on validation accuracy	Augmented dataset	94.69%
Classical Model 2	The same ResNet18 on the original folder; class-weighted loss; flip, rotation and brightness/contrast augmentation; 12 epochs in two stages	Original dataset	96.78%
Hybrid Quantum Model 1	Frozen ResNet18 features → Linear 512→8 with tanh → 8-qubit, 2-layer circuit → Linear 8→4 (circuit only, 4,188 trainable parameters)	Cfg B, same 600 test images	85.83% (515/600)
Hybrid Quantum Model 2 (Hybrid C)	Frozen features, standardised → 8 angles (π·tanh) → the same circuit, plus a classical branch Linear 512→64→4 beside it; the two sets of class scores are blended with a learned weight (about 0.48 on the circuit)	Cfg B, same 600 test images	88.33% (530/600)

Hybrid C combines the circuit path with a parallel classical branch. Both branches receive features from the same frozen encoder and produce four class scores.

4.2. CLASSICAL BASELINE

The baseline is an ImageNet-pretrained ResNet18 whose final layer is replaced by dropout and a 512→4 linear layer. Adaptation was progressive: first only layer4 and the head were trained, then layer3 as well, using smaller learning rates for the pretrained blocks than for the head. Training used AdamW, label smoothing in most runs, a plateau scheduler and early stopping on validation accuracy, restoring the best-validation checkpoint. EfficientNet-B0 was tried in Notebook 1 under the same budget and rejected because it performed much worse (62.75%, Section 5.1).

Classical Model 1 (Configuration B) uses an explicit grayscale conversion to three channels, dropout 0.3 and label smoothing 0.1; its best validation accuracy came at epoch 4 of 7. Classical Model 2 (Configuration C) uses all 6,400 original images with a class-weighted loss to compensate for the imbalance, adds rotation and brightness/contrast jitter to the flip, and trains for 12 epochs in two stages.

Checkpoint selection uses a consistent validation metric and retains the best checkpoint for test evaluation. Model selection and test reporting are kept as separate stages of the training procedure.

4.3. FEATURE EXTRACTION

After fine-tuning, the best ResNet18 is frozen and its classification layer is dropped. Each image is then converted once to a 512-dimensional vector and the vectors are stored. The shallow and quantum heads train on these tensors and not on images, which is what makes circuit simulation tractable: the convolutional part is run once and the circuit sees only small vectors. The same cached vectors feed the SVM, Hybrid Quantum Model 1 and Hybrid C, so the heads differ only in their own structure.

4.4. HYBRID QUANTUM-CLASSICAL CLASSIFIER

The hybrid model (Figure 4.1) maps a 512-dimensional feature vector to four class scores in four steps: a linear projection to eight angles, quantum encoding, the variational circuit with Pauli-Z readout, and a classical head. Each step is explained below together with the reason for the design choice.

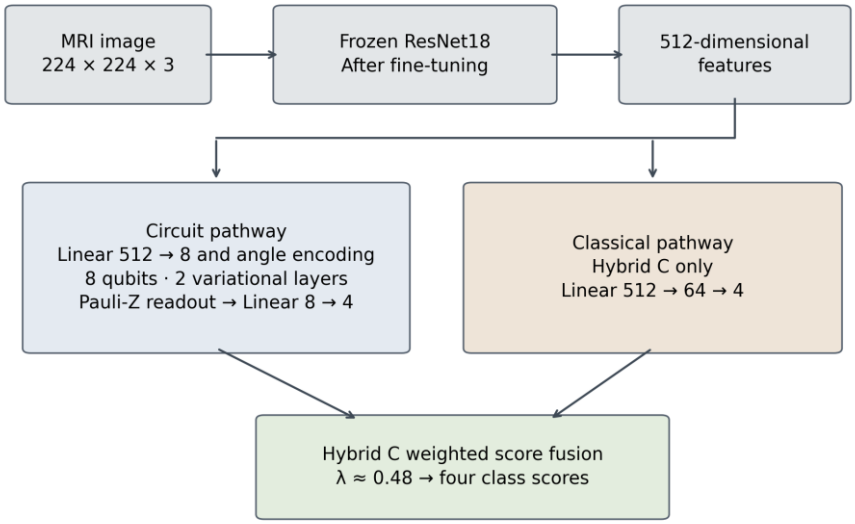


Figure 4.1: Architecture of the hybrid classifiers. Hybrid Quantum Model 1 uses the circuit path only; Hybrid C adds the parallel classical pathway.

4.4.1. Feature Projection

The classifier uses one rotation angle per qubit. A trainable linear projection maps the 512-dimensional feature vector to eight values, making the input compatible with the selected 8-qubit architecture. Later runs, including Hybrid C, first standardise features with training-set statistics:

$$z = W f + b, \quad z \in R^8 \tag{4.1}$$

This layer holds most of the head's parameters (4,104 of the 4,188 in Hybrid Quantum Model 1) and is learned jointly with everything after it. It is also the narrowest point of the model: everything the circuit can use has to pass through these eight numbers.

4.4.2. Qubit Encoding (Angle Embedding)

Each of the eight numbers is loaded into one qubit as a rotation angle. In the first runs (the 4-qubit run and Hybrid Quantum Model 1) the angles were bounded to $[-1, 1]$ radians by a tanh and applied with PennyLane's default X-rotation embedding (Equation 2.3). Later runs, including Hybrid C, scale the angles as $\pi \cdot \tanh$ so that they span $[-\pi, \pi]$ and use Y-rotations. Angle embedding needs one gate per qubit, makes the state vary smoothly with the input and adds no trainable parameters of its own.

4.4.3. Variational Circuit and Entanglement

The trainable circuit contains two StronglyEntanglingLayers (Figure 4.2). Each layer applies a three-angle rotation on every qubit followed by CNOT entanglers. PennyLane's layer template determines the connectivity; the figure illustrates the entangling structure

schematically. With L = 2 and n = 8, the rotation tensor contains 2 × 8 × 3 = 48 trainable weights. Entanglement allows measured outputs to depend jointly on the encoded inputs.

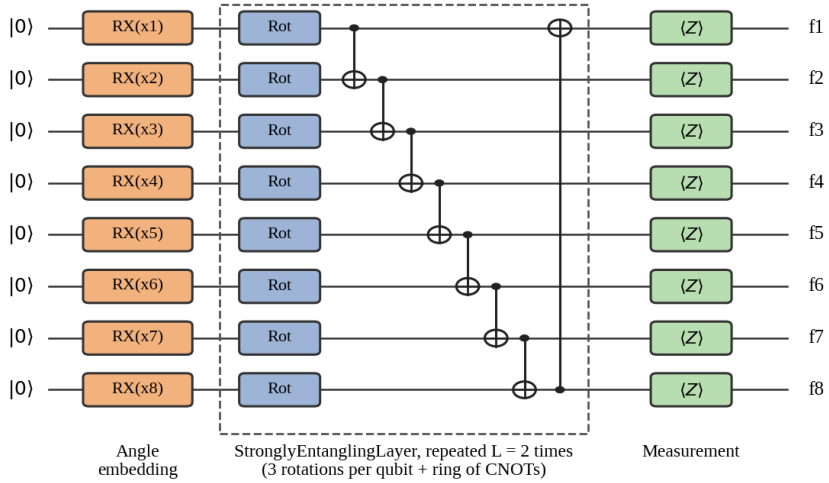


Figure 4.2: Schematic 8-qubit circuit with angle embedding, two trainable entangling layers and Pauli-Z readout. RX encoding is shown; later runs use RY.

4.4.4. Pauli-Z Measurement

The circuit is read out by measuring the Pauli-Z expectation value of each qubit (Equation 2.4), giving eight numbers between −1 and 1:

$$o_k = \langle \psi(z, \theta)| Z_k |\psi(z, \theta)\rangle \in [-1, 1], \quad k = 1,\ldots,8 \tag{4.2}$$

Gradients flow back through the simulator (by backpropagation or PennyLane's adjoint method), so the circuit trains like any other PyTorch layer. All circuits are simulated with exact expectation values and no measurement shots.

4.4.5. Classification Head and the Classical Branch of Hybrid C

In Hybrid Quantum Model 1 the eight circuit outputs go to a single linear layer, Linear(8, 4). Table 4.2 gives its parameter budget: 4,104 of the 4,188 trainable parameters sit in the classical projection and only 48 in the circuit.

Table 4.2: Trainable parameters of Hybrid Quantum Model 1

Component	Shape	Parameters
Projection	Linear(512, 8)	4,104
Quantum circuit	2 layers × 8 qubits × 3 angles	48
Output head	Linear(8, 4)	36
Total		4,188

Hybrid C adds a classical branch, Linear(512→64) followed by Linear(64→4), that sees all 512 features, beside the circuit path. The two sets of class scores are blended with a learned mixing weight, which stays near 0.48 for the circuit:

$$\hat{y} = \lambda \cdot \hat{y}_circuit + (1 - \lambda) \cdot \hat{y}_classical, \quad \lambda \approx 0.48 \tag{4.3}$$

The parallel classical branch contains 33,092 parameters: 32,832 in the 512→64 layer and 260 in the 64→4 layer. The mixing coefficient controls the weighted combination of the two logit vectors. A coefficient near 0.48 is a model setting, rather than a quantitative measure of the circuit's contribution to accuracy.

4.5. TRAINING AND IMPLEMENTATION

The models are implemented in PyTorch, with PennyLane providing the differentiable circuit layer. Table 4.3 summarises the recorded settings. Figure 4.3 shows the two-stage training arrangement: first adapt the encoder and classical classifier, then freeze the encoder and optimise the hybrid head on cached features.

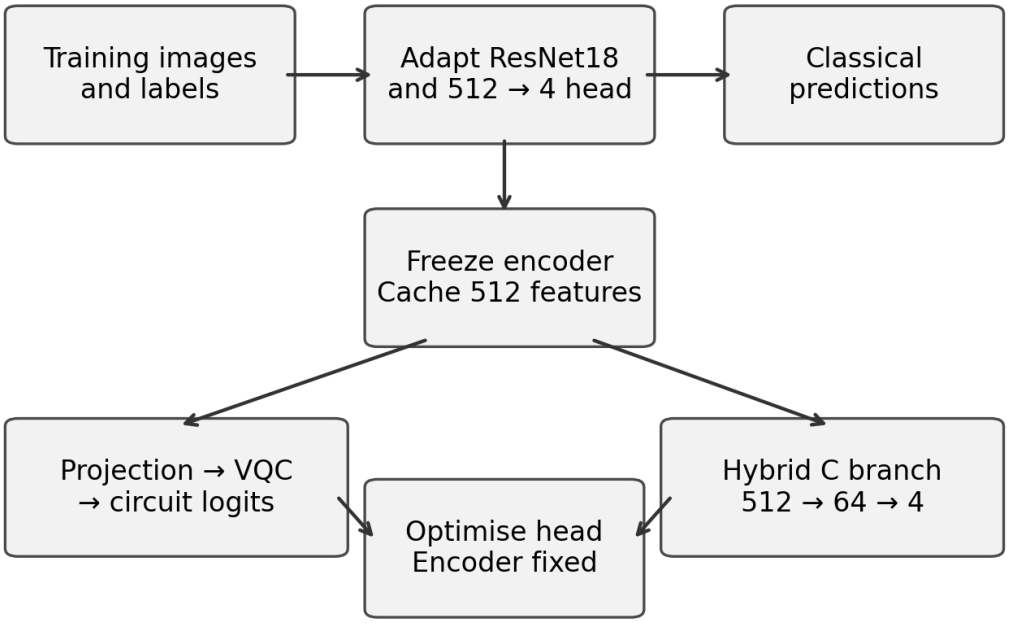


Figure 4.3: Encoder adaptation followed by frozen-feature hybrid head training.

Table 4.3: Recorded training settings of the four models

Setting	Classical 1	Classical 2	Hybrid 1	Hybrid C
Optimiser	AdamW	AdamW	Adam, lr 1e-3	AdamW, lr 5e-4

Setting	Classical 1	Classical 2	Hybrid 1	Hybrid C
Epochs	Early stopped; best val. at epoch 4 of 7	12, in two stages	10, no validation split	Stopped after 3; best val. at epoch 1
Loss and regularisation	Dropout 0.3, label smoothing 0.1	Class-weighted loss	Cross-entropy	Cross-entropy; classical branch 512→64→4
Input	Images, grayscale 3-channel	Images, flips, rotation, jitter	Cached 512-d features	Cached 512-d features, standardised
Seed	42 (split)	42 (split)	42 (split)	123 (training)
Compute	Tesla T4 GPU	Tesla T4 GPU	CPU simulator	CPU simulator

4.6. INTEGRITY CHECKS AND PROPOSED CONTROLLED EVALUATION

4.6.1. Integrity Checks Carried Out

The evaluation includes two complementary checks. StratifiedGroupKFold is used in the updated classical runs to keep assigned groups separate across folds. The earlier Notebook 2 audit examines identical feature vectors, nearest-neighbour cosine similarity, validation construction and an RBF SVM on the cached features. Section 5.4 presents those earlier audit findings separately so they are not mistaken for an audit of the updated folds.

4.6.2. Proposed Controlled Evaluation

Table 4.4 lists the follow-up experiments. Group-aware folding is already used in the updated classical evaluation; the next step is to apply the same grouping protocol to both model families and compare their components under matched conditions.

Table 4.4: Proposed controlled evaluation, in priority order

Step	What would be done	Threat it closes
1. Ablate the circuit	Train Hybrid C with the quantum branch removed, on the same split and seeds	Attribution of the 88.33% to the circuit
2. Split before augmenting	Use the same group identifiers for both families; keep all variants of a source within its assigned group and augment training data only.	Consistent leakage controls across the model comparison
3. Match conditions	Apply the updated StratifiedGroupKFold protocol to both families with identical folds and report macro-F1 beside accuracy.	No hybrid result on the original-data setting

Step	What would be done	Threat it closes
4. Independent validation	Cross-fit the features (each fold's features from a backbone that never saw it) or hold images out of backbone training	Validation at 100% cannot guide early stopping
5. Several seeds	At least five seeds; mean and standard deviation; paired test on the same test images	Single-run noise of about ±2.5 points
6. Like-for-like cost	Count parameters and time with the backbone included, on the same device	Misleading parameter and speed ratios

The follow-up decision rule should be fixed before testing: compare Hybrid C with its classical branch alone and a matched classical replacement for the circuit, using identical partitions and multiple training seeds. Report paired differences and their uncertainty. A circuit contribution should be claimed only when the observed improvement is supported by the predefined paired analysis and is consistent across repeated runs.

CHAPTER 5

RESULTS AND DISCUSSIONS

This chapter presents classical-head accuracies of 94.69% on augmented data and 96.78% on original data, together with the hybrid development results. The classical evaluation uses StratifiedGroupKFold. Hybrid metrics and audit findings refer to the earlier experiments, whose settings are described in Chapter 4.

5.1. CLASSICAL BASELINES

Table 5.1 summarises the classical experiments and the SVM control, in the order they were run. ResNet18 was kept as the encoder because EfficientNet-B0 stayed near 63% (it was trained mostly with a frozen backbone, and the notebook did not investigate why it lagged).

Table 5.1: Classical baseline development and final test accuracies

Model and change	Cfg	Validation	Test
ResNet18, layer4 + head trainable, early-stopped after 5 epochs	A	76.25%	79.50%
SVM (RBF, C=1, gamma 0.005) on that model's 512-d features	A	78.25%	79.50%
ResNet18, layer3 + layer4 + head trainable, 10 epochs	A	80.50%	81.00%
EfficientNet-B0, last two blocks trainable, 15 epochs	A	62.50%	62.75%
ResNet18, layer4 + head, 5 epochs	B	87.50%	87.00%
Classical Model 1: ResNet18, layer3 + layer4 + head, dropout 0.3, label smoothing 0.1	B	—	94.69%
Classical Model 2: ResNet18, layer3 + layer4 + head, class-weighted loss, flip/rotation/brightness augmentation, 12 epochs in two stages	C	—	96.78%

ResNet18 improved from 79.50% to 81.00% in the earlier Configuration A experiments, and an earlier Configuration B run reached 87.00%. The completed classical evaluation reaches 94.69% on augmented data and 96.78% on original data. The original-data result is 2.09 percentage points higher. Because the two datasets differ in class distribution and construction, this gap describes their experimental outcomes rather than the effect of a single model change.

5.2. CLASSICAL VERSUS HYBRID QUANTUM MODELS

Table 5.2 and Figure 5.1 compare the four principal models. Classical Model 1 is the augmented-data classifier and Classical Model 2 is the original-data classifier, both evaluated with group-aware

folding. The hybrid scores come from the earlier Configuration B experiments. This distinction matters when comparing accuracy across the two stages of the project.

Table 5.2: Final results comparison

Model	Cfg	Test samples	Test accuracy	Correct predictions	95% Wilson CI	Macro-F1
Classical Model 1 (ResNet18)	Aug.	—	94.69%	—	—	—
Classical Model 2 (ResNet18)	Orig.	—	96.78%	—	—	—
Hybrid Quantum Model 1 (8-qubit head)	B	600	85.83%	515/600	82.8–88.4%	0.86
Hybrid Quantum Model 2: Hybrid C	B	600	88.33%	530/600	85.5–90.7%	0.8821

The table reports the metrics available for each run. Classical test accuracy is available for the updated evaluation; its test support, correct-prediction counts and macro-F1 are not reconstructed. Hybrid counts, intervals and macro-F1 are retained from the earlier outputs. A dash means that the corresponding statistic is not available.

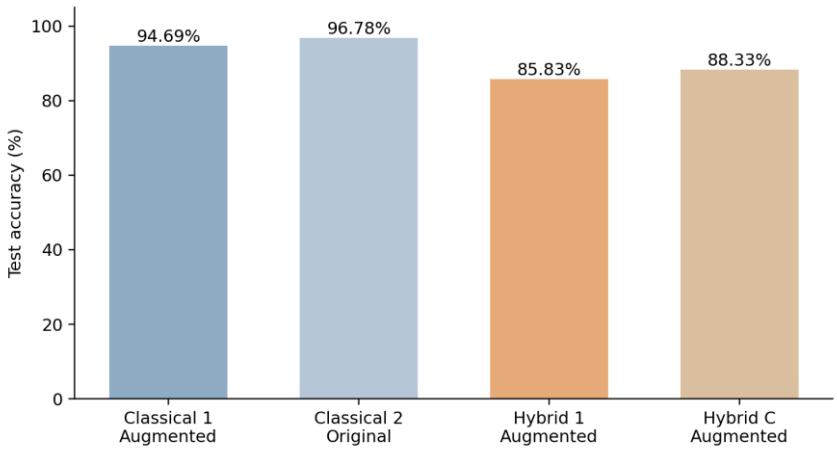


Figure 5.1: Test accuracy of the classical and hybrid models by dataset.

The original-data classical head gives the highest accuracy at 96.78%. The augmented-data classical head reaches 94.69%, compared with 88.33% for Hybrid C and 85.83% for the circuit-only hybrid. Hybrid C's macro-F1 is 0.8821, with an inferred count of 530 correct predictions out of 600.

The augmented classical head is 6.36 percentage points above Hybrid C and 8.86 points above the circuit-only model. These scores favour the classical head in observed accuracy. They span different evaluation stages, so they should not be treated as a paired

test on identical folds. A stronger architecture comparison would train both model families with the same group identifiers, folds and selection procedure.

Table 5.3: Per-class F1 of the hybrid models

Class	Hybrid C (B)	Hybrid 1 (B)
MildDemented	0.91	0.89
ModerateDemented	0.98	0.97
NonDemented	0.84	0.81
VeryMildDemented	0.79	0.77
Macro F1	0.88	0.86

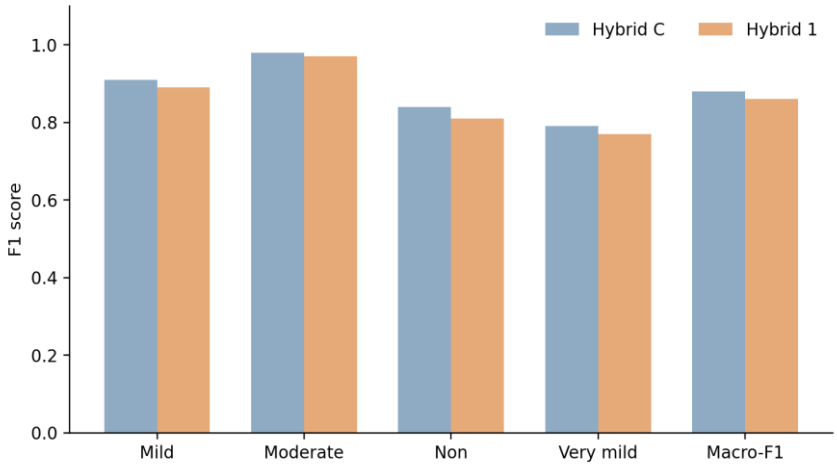


Figure 5.2: Per-class F1 of Hybrid C and Hybrid Quantum Model 1.

VeryMildDemented is the weakest class for both hybrid models: F1 is 0.79 for Hybrid C and 0.77 for the circuit-only head. Their ModerateDemented scores are 0.98 and 0.97. The normal–very-mild boundary therefore deserves attention in subsequent hybrid experiments. These per-class scores belong to the hybrid runs and are not assigned to the updated classical models.

Table 5.4 reports computational measurements from Notebook 2. ResNet18 contains 11.18 million total parameters and 10.50 million trainable parameters. The circuit-based head contains 4,188 trainable parameters in addition to the frozen encoder. Recorded inference times are 3.21 ms per image for the GPU image model and 10.78 ms for the CPU circuit head on cached features. These

measurements describe the implemented paths; they use different devices and input boundaries. Reported weighted one-vs-rest AUC values are 0.978 and 0.950, respectively.

Table 5.4: Recorded computational measurements from Notebook 2

Quantity	Final ResNet18	8-qubit hybrid head
Trainable parameters	10.50 million (11.18 million in total)	4,188 (excludes the frozen backbone)
Inference time per image	3.21 ms (GPU, from pixels)	10.78 ms (CPU simulator, cached features)
Weighted one-vs-rest AUC	0.978	0.950

5.3. DEVELOPMENT OF THE HYBRID MODELS

Table 5.5 and Figure 5.3 summarise the recorded hybrid development runs in execution order. The pilot reports validation accuracy; subsequent entries report test accuracy. The reference line shows the latest augmented classical-head accuracy of 94.69%. All circuits run on a noiseless state-vector simulator.

Table 5.5: Hybrid quantum runs in the order they were tried

Run	Circuit and classical parts	Training	Result
Pilot (Notebook 1, Cfg A)	8 qubits, 2 layers; 512→64→8 projection; head 8→4	AdamW 1e-3, batch 8, one 60 s pass over 1,552 of 3,200 samples	Validation 65.00%; NonDemented never predicted
4-qubit run (Cfg B)	4 qubits, 1 layer, 12 weights; 512→4 tanh; head 4→4	Adam 3e-3, 3 epochs, batch 16	Test 66.83%; VeryMild recall 0%
Hybrid Quantum Model 1 (8-qubit first run)	8 qubits, 2 layers; 512→8 tanh; head 8→4	Adam 1e-3, 10 epochs, no validation split	Test 85.83%
8-qubit with validation split	Scaled features, $\pi\cdot\tanh$ angles, head 8→16→4, class weights	AdamW 3e-4, 8 epochs, 15% of training set held out	Test 83.33%
After removing duplicates	Same architecture	Adam 1e-3, 6 epochs, 12 training rows removed	Test 72.17%
Hybrid A	Same circuit; 512→8 linear; head 8→4	AdamW 5e-4, gradient clipping, 5 epochs	Test 68.17%
Hybrid B	Adds classical branch 512→32→4; class scores added	As Hybrid A	Test 87.83%

Run	Circuit and classical parts	Training	Result
Hybrid C	Classical branch 512→64→4 and a learned mixing weight (≈0.48 for the circuit)	Seed 123, AdamW 5e-4, stopped after 3 epochs	Test 88.33%

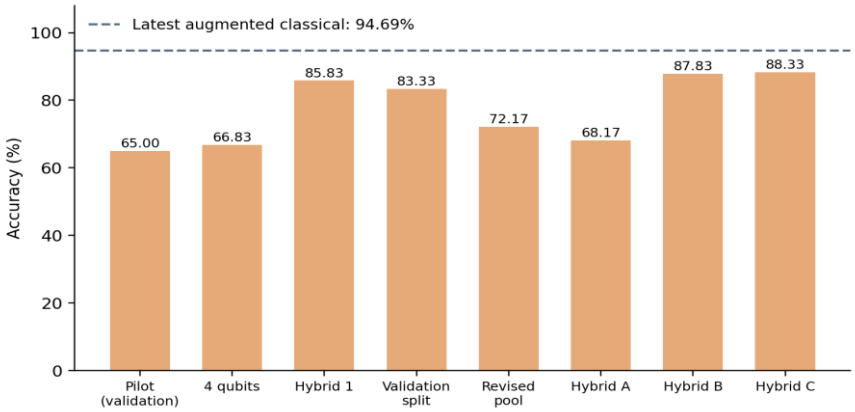


Figure 5.3: Accuracy across hybrid development runs. The pilot uses validation accuracy; other bars use test accuracy. The dashed line is the augmented classical score of 94.69%.

Circuit-only test accuracies range from 66.83% to 85.83% across the recorded recipes. The pilot was budget-limited and never predicted NonDemented, while the four-qubit run recorded zero VeryMildDemented recall. Adding a classical branch produced 87.83% for Hybrid B and 88.33% for Hybrid C. These runs describe architecture development rather than a controlled ablation. The latest augmented classical accuracy of 94.69% exceeds every recorded hybrid result.

Hybrid C gives the strongest hybrid result. Its parallel branch has access to all 512 encoder features, while the circuit pathway receives eight projected angles. Comparing the fused model with each branch alone would show whether combining these pathways improves classification.

5.4. WHAT THE INTEGRITY CHECKS FOUND

The following findings belong to the earlier Notebook 2 experiments. They motivated a closer examination of data separation. The updated classical evaluation uses StratifiedGroupKFold; the earlier audit is retained here to explain the development of the evaluation protocol.

1. Identical feature vectors: 12 train–test pairs cover six distinct test images, all labelled ModerateDemented. The corresponding 12 training rows were removed,

reducing the training pool from 4,800 to 4,788 rows. Identity of extracted vectors is the recorded check; it is not an independent raw-image hash audit.

2. High feature similarity: 1 of 600 test vectors has nearest-training cosine similarity at least 0.999, 36 at least 0.99 and 91 at least 0.95 (Figure 5.4). High cosine similarity indicates close learned representations; it does not by itself establish that two source images are duplicates.

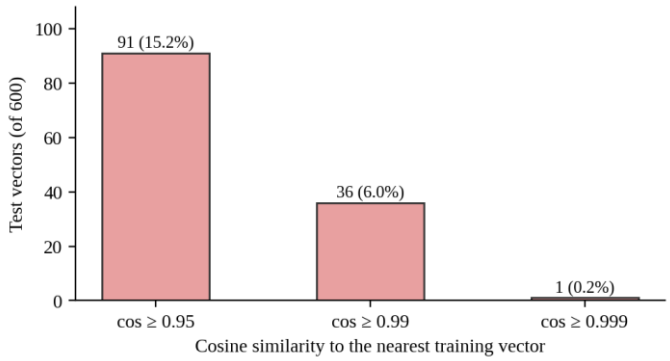


Figure 5.4: Counts of test vectors above each nearest-training cosine-similarity threshold.

3. Head validation construction: the backbone records approximately 99.98% training accuracy. The hybrid validation slice was drawn from features of images previously used to fit that backbone. Hybrids B and C therefore reached approximately 100% head validation accuracy while their test accuracies were 87.83% and 88.33% (Figure 5.5).

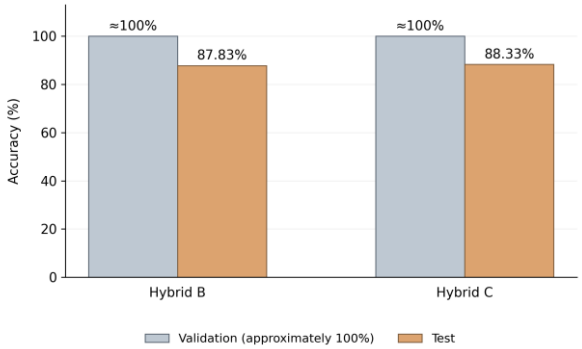


Figure 5.5: Reported validation and test accuracy for Hybrids B and C.

4. Revised-pool run: after removing the identified training rows, one run reached 72.17%, compared with 83.33% for an earlier recipe. Learning rate, class weighting and epoch count also changed. The score difference therefore cannot isolate the effect of removing repeated vectors.

5. SVM check: the Notebook 2 RBF SVM records 100% validation accuracy but 25.0% test accuracy and macro-F1 of 0.10. The earlier Configuration A SVM

records 79.50% test accuracy. The later discrepancy remains unresolved and is excluded from conclusions about circuit performance.

These checks show why separating related samples matters, particularly for the small original ModerateDemented class. Group-aware folding addresses the partitioning concern in the updated classical evaluation. Validation independence and the contribution of each hybrid branch remain separate questions for a matched follow-up experiment.

5.5. WHY EACH RESULT CAME OUT AS IT DID

Three aspects help explain the results: adaptation of the encoder, compression of its features and the parallel classical pathway. The observations below suggest useful follow-up experiments without assigning a cause that the current runs have not isolated.

Encoder adaptation. Progressive unfreezing and the larger balanced subset coincide with improved classical performance. ResNet18 provides a common 512-dimensional representation for subsequent heads. Since the data configuration and regularisation also changed during development, the observed improvements should be interpreted as gains in the complete training recipe.

Original-folder performance. The latest classical head on the original dataset achieves 96.78%, the strongest reported result. The augmented-data classical head achieves 94.69%. The original-folder and augmented-folder experiments describe different data distributions, so their 2.09-point gap should be interpreted as a dataset-specific outcome rather than an isolated architecture effect.

Feature compression. The circuit-only hybrid projects 512 features to eight angles before producing eight measured outputs. This creates a compact trainable interface, with most head parameters in the projection. Its 85.83% test accuracy establishes that this representation supports the task, while a matched eight-dimensional classical head would help determine whether the circuit improves that representation.

Parallel feature pathway. Hybrid C retains access to the full feature vector through its 512→64→4 classical branch. Its 88.33% accuracy exceeds the circuit-only hybrid by 2.50 percentage points but remains 6.36 points below the latest augmented classical result. Branch ablation is needed to quantify how each pathway affects predictions.

Computational accounting. Feature caching reduces repeated encoder computation while fitting shallow heads. The small trainable circuit head remains dependent on the frozen ResNet18 encoder for inference from MRI pixels. A complete deployment

comparison should therefore report encoder plus head parameters and end-to-end timings on a common device.

5.6. DISCUSSION

The classical heads achieve the highest accuracies: 94.69% on augmented data and 96.78% on original data. StratifiedGroupKFold makes separation of assigned groups part of the updated evaluation. The hybrid runs show that a simulated circuit can learn from ResNet18 features, although both hybrid variants remain below the augmented classical result. The next comparison should bring the hybrid models onto the same group-aware evaluation protocol.

For the four-class task, overall accuracy should be read together with VeryMildDemented performance. The relatively low F1 of that class indicates that improvements should focus on the normal–very-mild boundary. The next comparison should preserve the data split and encoder while changing only the head, which would make architecture effects easier to assess.

Table 5.6 summarises the main findings and their practical interpretation.

Table 5.6: Main findings and interpretation

Finding	**Interpretation**
94.69% classical vs 88.33% hybrid on augmented data	The classical score is 6.36 points higher; the runs use different evaluation stages.
85.83% for the circuit-only head	Eight encoded angles and a 48-weight circuit support four-class classification.
96.78% on the original dataset	Highest reported classification accuracy in the revised comparison.
Hybrid VeryMildDemented F1 of 0.77–0.79	The earliest labelled stage is the principal class-wise challenge.
4,188 trainable head parameters	Compact adaptation layer; the frozen encoder remains part of the complete model.

5.7. LIMITATIONS AND THREATS TO VALIDITY

Table 5.7 summarises the remaining evaluation considerations. It distinguishes the group-aware protocol now in use from the checks still needed for a matched comparison.

Table 5.7: Evaluation boundaries and required controls

Boundary	Recorded evidence	Required control
Group definition and separation	StratifiedGroupKFold is used in updated classical evaluation; earlier feature audits found related representations across splits.	Document group identifiers and fold membership; keep source variants in one group and audit image hashes.
Hybrid validation	Head validation images were previously used in backbone fitting.	Hold validation images out of encoder training or use cross-fitted features.
Circuit attribution	Hybrid C includes a full-feature classical branch; no isolated branch control.	Compare circuit-only, branch-only and fused heads under matched settings.
Selection and uncertainty	Several development runs reuse the test set; results are single runs.	Fix selection on independent validation; report repeated seeds and paired prediction analysis.
Cost and execution	CPU simulated head on cached features versus GPU encoder from pixels.	Measure end-to-end cost with the backbone included and a common timing boundary.
Generalisation	Public 2-D JPEG slices; earlier Configuration C test set had nine ModerateDemented images.	Evaluate independent cohorts and report class supports; study hardware noise separately.

CHAPTER 6

CONCLUSIONS AND SCOPE FOR FURTHER WORK

6.1. CONCLUSIONS

We compared classical and hybrid quantum-classical models for four-class Alzheimer MRI classification. The classical head achieves 94.69% on augmented data and 96.78% on original data using StratifiedGroupKFold to reduce leakage between assigned groups. The earlier circuit-only hybrid reaches 85.83%, while Hybrid C reaches 88.33% with macro-F1 of 0.8821. The original-data classical head is the strongest model in the reported experiments.

The augmented classical result is 6.36 percentage points above Hybrid C and 8.86 points above the circuit-only hybrid. The small circuit head is trainable, but it still depends on the pretrained encoder and does not show an accuracy advantage here. A matched group-aware evaluation and branch ablation would provide a clearer account of the circuit's contribution. The project establishes the complete pipeline and gives a practical direction for improving its hybrid component.

6.2. SCOPE FOR FURTHER WORK

Future work, in priority order, is to:

- ablate the circuit by training Hybrid C with the quantum branch removed, on the same split and seeds;
- extend StratifiedGroupKFold to a matched classical-versus-hybrid evaluation, audit group membership and study an independent dataset where possible;
- run both families under the same conditions (Configuration C), report macro-F1 and repeat with at least five seeds with paired tests;
- make validation independent by cross-fitting features, and resolve the unexplained SVM result;
- compare parameters and time like for like, vary qubit count, depth and embedding against matched classical controls, evaluate under simulated noise and, if possible, on hardware, and add Grad-CAM [15] and per-class clinical analysis.

6.3. REPRODUCIBILITY

The earlier development splits use seed 42, and Hybrid C training uses seed 123. The updated classical evaluation uses StratifiedGroupKFold. Implementation uses PyTorch 2.11, PennyLane 0.45.1 and a Tesla T4 GPU for CNN training. Reproduction requires the notebooks, group identifiers, fold assignments, class mappings, checkpoints and prediction files. Reference [1] identifies the dataset.

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APPENDIX A
COUNTS, CONFIDENCE INTERVALS AND PARAMETERS

This appendix gives count calculations for the earlier hybrid runs and parameter calculations for the model components. Classical accuracies are reported in Chapter 5. Their prediction counts are not inferred from percentages alone.

Table A.1: Correct-prediction counts for the recorded hybrid runs

Model	Test images	Reported accuracy × n	Count	Check
Hybrid Quantum Model 1	600	0.8583 × 600 = 514.98	515	515/600 = 85.83%
Hybrid C	600	0.8833 × 600 = 529.98	530	530/600 = 88.33%

The hybrid counts above are obtained by rounding accuracy multiplied by test-set size. They agree with the displayed percentages and apply only to the earlier hybrid evaluation.

Table A.2: Derivation of parameter counts

Model or part	Calculation	Total
Hybrid Quantum Model 1 head	(512×8 + 8) + (2×8×3) + (8×4 + 4) = 4,104 + 48 + 36	4,188
Hybrid C classical branch	(512×64 + 64) + (64×4 + 4) = 32,832 + 260	33,092
Final ResNet18	As reported in Notebook 2	11.18 M (10.50 M trainable)

The Hybrid C branch count follows from the layer sizes in the notebook description; the notebook output does not list a total for Hybrid C.

The 4,188-parameter hybrid head comprises a 4,104-parameter projection, 48 circuit weights and a 36-parameter readout. Circuit weights account for about 1.1% of the head. The head-to-trainable-backbone ratio is approximately 0.04%, but the encoder remains required. For Hybrid C, 530 correct classifications and n = 600 give a nominal Wilson interval of approximately 85.5–90.7%.

APPENDIX B
PLAGIARISM REPORT

The institutional similarity certificate is a separate submission requirement. No similarity score or certificate was supplied with the source document; this revision does not create one.